

Senior Project Portfolio

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Williams

ETHICS PAPERS

COINS

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Is Using Broadband Internet Obtained From Someone Else's Open Router Ethical

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**COINS 325
Professor Roberts
February 2, 2006**

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IS USING BROADBAND INTERNET OBTAINED FROM SOMEONE ELSE'S OPEN
ROUTER ETHICAL?

Some would have you believe there is no such thing as absolute right or absolute wrong, but those people would be mistaken. Most of society can agree that child molestation, rape, ruthless murder and other such transgressions are wrong and wrong absolutely. However with technology advancing as it, the law of man finds itself previously inadequate to establish right and wrong by tradition standards when dealing with issues about computer and/or internet ethics. The era of the internet has brought up numerous new moral issues, forming a gray area in what once was considered to be mostly black or white. Even crimes that have been committed since the beginning of time are suddenly made more complex morally as they are advanced by technology. Even old felonies like robbing a bank, which is now doable with the internet, are being treated differently in courts of law when there is computer involved. While robbing a bank is still illegal, the punishments for those who are caught trying to hack into bank's account are often less than those who held up a bank using guns, masks, and a bag. (One argument has been that it is because no one was at risk for being hurt.) Other moral dogmas have arrived, and trialed by society, such as the infamous Napster debate, which forced some to ask the question *is downloading free music a crime?*

Somebody once said nothing in life is black and white, while others have said some issues are clearly right and others are clearly wrong. Whether using broadband Internet obtained from someone else's open router is not one of those issues for most of America. Now this moral enigma is a complex one. Is a "crime" where nothing tangible is stolen and no one is hurt really a crime?

Some people have called using Internet that has been acquired from someone else's router unethical stealing. A great deal of this debate comes down to the question what is stealing? Stealing has been defined by Merriam-Webster Online Dictionary as "to take the property of another wrongfully and especially as an habitual or regular practice" or "to take surreptitiously or without permission". While it could be said one

who used another person's Internet once will do it again and eventually make doing so a regular practice, did they take the property of another? It is obvious did they did not take any tangible property, but in the computer world emphasis ceases to be about what is touchable. Was property stolen? In this case, under the assumption that Internet is property, at least two separate and opposite views can be formed, each one claiming to have just as much, if not more, validity as the next. View #1 is that customer who are paying for the product, as well as the as the company that provides the service, are being stolen from because someone who did not pay for it is freeloading off of both. In this view, one is supposed to view the free loader as a bulgur who breaks into a house and eats the owner's food, using the owner's shower, and steals owner's clothes and money. View #2 could be that one could not have stolen from the owner if that owner stills have the service available to him. This view shows the freeloader as a visitor who washed his hands without asking and then was accused stealing water.

There are many views on this matter, although have more legality than others. Some aspects of issue are clearly defined as wrong. If someone buys a wireless card and signs a contract in which that person agrees to never take the internet from a router other than his own, then to that person it is a crime. According to www.boingboing.net, those officials who feel that it is unethical to use Internet acquire from another router feel strong enough to act on it. "Benjamin Smith III, 41, faces a pretrial hearing this month following his April arrest on charges of unauthorized access to a computer network, a third-degree felony. Police say Smith admitted using the Wi-Fi signal from the home of Richard Dinon, who had noticed Smith sitting in an SUV outside Dinon's house using a laptop computer." BoingBoing's opinionated Xendi Jardin wrote "Look, I know it's not quite this simple, but -- an open network? Perhaps next they'll arrest folks for having the audacity to walk down the street and breathe their neighbors' air."

In all honest, this issue cannot be realistic debated in the format of this paper, because this piece neglects a few issues of great importance to this topic. Although using broadband borrowed from another's open router seems rather harmless, one must consider that same person could also be looking at files or performing potentially dangerous actions on other person's computer. Surfing the 'net gotten from someone's

open routing could so many more issues than web bowering. Those whose open routers are being used by an outsider are at high risk of being a victim of identify fraud as well as viruses and other unpleasanties. People could breaking into somebody bank account, or find the person's address and social security number. For these and other potentially dangerous reason, it would in the best internet of paying customers if it was not legal to use broadband off of someone else's open router. According www.sptimes.com, with reports the same newspaper story as www.boingboing.com with a very different bias, people who use others' routers purpose a threat for the safety and security of the router's owners. "A drive through downtown St. Petersburg shows how porous networks can be. In less than five minutes, a *Times* reporter with a laptop found 14 wireless access points, six of which were wide open. "I'll guarantee there are tons of people out there who have their wireless network being exploited but have no idea," Breeden said. "And as we see more people utilizing wireless, we'll see more people being victimized."

Those who only use internet taken from another's open router and do not access the owner's computer claim they are doing nothing wrong are still borrowing internet without the owner's permission and against the broadband internet providing company's policy. The company loses money which results in that company charging more for internet service to make up for lost profits.

After all the augments has been made and all the pros and cons have been listed, the truth still remains. Using broadband Internet obtained from someone else's open router is simply unethical. Those who do so are ripping off the Broadband companies. They are also providing themselves a temptation to look into the customer's private files and potentially other crimes. Basically, if one did not pay for it and it is not free, then that one should not have it.

Work Cited Page

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Both sides of the coin:
A look at the RIAA and piracy

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14 November 2007

Both sides of the coin:
A look at the RIAA and piracy

In this digital age, most of mankind praises technology for the great many advances of this generation, while cursing the same technology for its complications. Out of all of modern technology's complexities, however, legal issues are perhaps the most troublesome of all. No group, however, seems to be having as hard a time dealing with these changes as does the Recording Industry Association of America (RIAA). The RIAA deals with, among other things, the issue of piracy, often making news headlines. In this brave new world, the RIAA is trying to find a solution to what it considers brave, new crimes. For situations such as the ones with which the RIAA deals, there are no absolute clear answers. Defining right and wrong is beyond challenging, and to do so would be unrealistic. The RIAA is not totally right. The college students are not completely wrong. Choosing one side over the other would be a mistake, mostly based on personal bias. Instead, this ethic paper will examine both sides of the issues from several points of view.

The article given was too biased to really use as a legitimate source. Even to this final sentence, the article uses unwarranted terms, such as "the sad part," and words like "ridiculous", which have such a partiality that they actually prevent the readers from having purely objective consideration. The article is even so bold to portray the RIAA as inhumane lawmakers who care about their music sales more than the well-being of drivers and passengers by stating "the music industry considers the tackling of music piracy as far more important than protecting people's lives on the road". Whether or not the article expects its readers to be so enraged at the RIAA that they blindly agree with any opposing force or if it believes its audience is highly lacking in intelligence is unclear, but it is pretty clear that the intended audience is not the sound of mind. The RIAA policies and laws clearly only deal with the recording industry, and, that being the case, have very little, if any, influence on the amount fined in a speeding ticket. To imply that there is any connection between RIAA's lawsuits and DMV's traffic laws is greatly ludicrous. Just because RIAA's lawsuits are more expensive and have a higher 'conviction' rate, does not mean they think lightly of drivers' and passengers' safety. The article, which is nameless so this paper will continue to refer to it as 'the article', might even do better to ask the question "if the RIAA takes piracy so seriously, why does our government not punish speeders more severely?"

According to the article, student Sarah Barg from University of Nebraska-Lincoln knows "she done wrong", and yet still considers the approach of the RIAA as "bullying". Dictionary.com defines a bully as "a blustering, quarrelsome, overbearing person who habitually badgers and intimidates smaller or weaker people", so one can definitely appreciate her word choice. The RIAA could deal with the situation in less aggressive ways, or at least send her a warning notice before they sued her. Without doubt the RIAA has the aggressive, overbearing and intimidation qualities of a bully, which is not the same thing as being unjust or wrong. Simply because they are hostile does not necessarily mean that they are immoral. Do the thieves have a right to expect nice treatment from the person they have stolen from? Would you blame anyone you know for being less-than-understanding to a bandit after being mugged?

According to an Encyclopædia Britannica article entitled "Cybercrime: File sharing and piracy", the RIAA sees the issue as simply wrong or right. In their opinion the college students did not pay for the music and therefore they have no right to process it. Users of peer-to-peer services might view it differently. There was at least one user who brought the CD, so he or she owns it and therefore has the right to share his or her property. Even if some members of the RIAA agree with that argument, those members would not agree that those same users have a right to make a music CD available to millions of people worldwide, giving them the ability to create or even mass produce their own illegal CDs. The debate becomes less about if users have a right to share copyrighted and trademark material for which they paid, and therefore, legally obtained. The debate becomes about the consequences of sharing. Where is the line from whence sharing becomes stealing? The RIAA believes that such file-sharing services were responsible for the creation of a new breed of criminal: a digital pirate. The RIAA is of the mindset that where there is a crime, the must, for the

necessary of order and justice, be a punishment.

When examining moral issues, one must realize that good motives or intentions do not necessary produce good results. Although the RIAA will claim they only desire justice, there actions are often looked at as if they were extreme or even I unjust. Suing college students and threatening lawsuits seems, as Sarah Barg might phrase it, bully-ish. In the RIAA's haste for what they consider justice, they may have retorted to unprofessional methods. Lawsuits by e-mail hardly seems efficient. The RIAA might be its own worst enemy if it continues to use methods that undermine the good they are trying to do.

The article states "students don't realize the danger of file sharing until they get caught, since no e-mail has been sent out to students warning about the RIAA's action and that threats alone are not going to solve the problem." If the RIAA sent out notices or warnings, I think that might go a distance to restore their image. On the other hand, one has to acknowledge the fact that the RIAA is under no legal standard to show courtesy to a guilty party. The police may give a person a warning if drive other the speed limit once or twice, but never a hundred times. Either way, whether a student downloaded one song or one million songs the same crime was committed and why should a student have to be warned for something they already knew is wrong.

The article also states "over 60 college campuses were targeted according to an RIAA spokeswoman and so far at least 500 university students in the US have paid up. By suing students, it is quite clear that students are going to have little other choice than to settle the hefty fee, since very few, if any, can afford to fight". The RIAA's methods, as unprofessional as they are, seem to be working a little, so the RIAA may feel it has little reason to stop using such immature tactics. The fact is, legally the RIAA is proven right over such file-sharing services such as Napster. The RIAA has never seemed to care about the public image, so they do not even attempt to appear nice or even understating to the millions of people who has downloaded a music track. If the RIAA continues to have the courts side with them in legal proceedings, moral or not the RIAA will continue to be bully, suing whoever they want, like Johnson, a college student who only downloaded a hundred songs.

Issue concerning file-sharing and music downloading are complex. It seems that money is motivator for all sides. The RIAA does not want the everyday person to download songs for free because the artists (not to mention the RIAA) do not receive any funds for their product. The users of Limewire and other such file-sharing services are probably encouraged to continue to use them by their desire to save money. With greed being a factor on most, if not all, points of view, it is a little hard to properly identified what is truly right. If one had to choice sides however, legally the RIAA may be considered to be more righteous then the groups and people who oppose it. Although the RIAA methods are lacking diplomatic cause, civil manner, tactful approach and intelligence delivery, their main goal is to make sure artists get wages from their work and there is nothing wrong with that.

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COINS-330-01

COMPUTER ARCHITECTURE

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JAVA PROGRAMMING
II
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COINS 326

14 September 2006

“An Analysis of the CSU Code of Conduct as it Relates to the Internet”

Most, if not every, college and university have a code of conduct that specifically addresses student computing and internet policies, and Charleston Southern University (CSU) is no exception. CSU's code of conduct, which includes twelve guidelines and articles about privileges, can be found on its website. In it there are a number of wonderful policies that every school should have as well as guidelines and rules that need improvement.

One of Charleston Southern University's wonderful rules in its code of conduct is the sixth guideline. This guideline reads “I accept the responsibility for all material received or sent through/over the network.” This is great guideline because instead of telling users what they can't or can do, it merely makes them that they are taking accountability. This is better than just having rules that specifically informs student that they can't do this or that, because humans have a tendency to rebel and do something simply because they are told not to do it. But this policy allows the user to chance to do the right thing simply because it is the right. Also, this guideline benefits the university by taking some of the reasonability from itself and placing it on the students.

One of CSU's policies that could use improvement is the ninth guideline. Guideline 9 states "While in the library/computer labs [the user] will not alter or attempt

to alter a computer setup or system configuration." (CSU) A better option would be for CSU to allow students to have limited ability to make minor alterations to a computer's set up. This change in rules would be valuable because there are many changes that would be educationally beneficial, not mention time saving, for the student to be able to make. For example, in the spring of 2006 in Java I, many students were unable to run their Java programs due to the fact that the preset path was incorrect. If students were allowed to change paths, this would have simply been a small inconvenience, and easily correctable by the students themselves. Instead, because changing the path require the user to have an administer password, the professor had to do the time consuming process all by himself.

Other Christian Colleges, such as Liberty University, are more relaxed about their code of conduct. Liberty's online policy consist of one three short paragraphs; the first informs to obey copyright right laws and not to not copy university-owned computer software", "which could as easier be argue that it was written because of their greed. (Liberty) The second paragraph informs one that there is a physical copy of all the rules under the code of conduct, but makes no mention on how to obtain it. The third paragraph warns students not to perform illegal activities and threatens that violation of these "standards will make a student subject to disciplinary action." (Liberty) Compared to CSU's online version, theirs seems inadequate. The word internet is not even mention under "Computing Ethics". Which I suppose is better than UNC, because that college did not include of code of conduct which relates to the internet at all.

On the other hand, other colleges have significantly more internet policies that of Charleston Southern University. The University of Southern California uses ITS

(Information Technology) Computing Policies, which is more than 10 times longer than CSU's code of conduct. (USC) Yale University is also governed by ITS Policies which again are extensive, but are slightly different than USC's. (Yale)

Some rules in the code of conduct of CSU are quite questionable. For example, under CSU's code of conduct the fourth guideline reads "I will not download or share copyrighted material of any kind including but not limited to songs, movies, television shows and software." (CSU) Does this mean only when it is downloaded illegally? Or is legitimately purchased music, like buying them from iTunes, included in that guideline as copyrighted material? If this instruction includes legally downloaded music, then what is the logic behind it? Student should be able to lawfully acquire any music they desire.

Although the majority, if not all, universities and colleges do process a code of conduct that particularly addresses student internet and computing policies, Charleston Southern University has a unique view on some of issues as shown above. Some aspects of the code could be improved, but nothing written by man is perfect. There are also some excellent guidelines when compared to the rules of some other colleges. Overall, the CSU code of conduct is more good than bad and they work better when everyone keeps them.

COINS 431 62

ADVANCE OPERATING
SYSTEMS & LAB

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COINS 431

24 September 2006

Video Game Violence: What is done and should be done

Exposure to violent games increases physiological arousal, increases aggressive thoughts and emotions. (Anderson & Bushman, 2001) "In a study of 8th and 9th graders, students who played more violent video games were also more likely to see the world as a hostile place, to get into frequent arguments with teachers, and to be involved in physical fights". (Gentile et al., under review) Violence video games negatively affect the player. (Anderson and Dill, 2000) Assume one knew those previous statements that one might wonder what is currently done about violence in video games. While there is not much being done about it by video game companies, there are some companies who are trying to deal with this issue. For example, in response to complains about brutal video games and the marketing thereof, the Digital Software Association has set up the Advertising Review Council (ARC), which began operations on January 31, 2000. ARC has developed a set of laws of marketing and advertising principles with mechanisms for enforcement. ARC, as well as other councils and organizations, has responded to the issue of marketing Mature-rated games to youth limiting ads for M-rated games in popular teen media and nearly always providing rating information in advertising. (Walsh, 2001) Also, people not directly working with video game are actively trying to change the method in which violent games are dealt with as well. On Tuesday, September 19 2006 in Fairfax, Julie Benay, associate principal of Mary S. Babcock Elementary School in Swanton finished with a proposal for legislation that imposes a user fee on violent and explicit video games, specifically those games rated by Entertainment Software Rating Board as being either Adults Only (AO) or Mature (M). (Burnham, 2006) However, realistic and legally there is not much that can be done about video game violence. This is because the animated content in video games has been considered protected speech under the First Amendment; there is an extremely narrow range of permissible government involvement with their advertising and marketing. (Carless, 2006)

There are certain things that should be done which should be done about violence in video games. One of the main actions that should be taken is making sure video game ratings are enforced. The 'M' rating stands for mature, meaning only people 18 and older are allowed to play it. In reality, however, all that rating means that only people over seventeen should be allowed to buy it. There are children who have their parents purchase the mature games. Others play mature games at friends' houses or over the internet, where all a minor has to do is lower his/her date of birth on the game site so that he or she is over 18. There needs to be a better system to prevent children from playing games designed for adults. Also, there needs to be some type of penalty for the parents or responsible adult who allow to their child to play games unlawful for them.

Another thing that would have should be done about the violence in video games is that the producers should be held partly responsible for negative aftermaths that are a result video games. John Bruce "Jack" Thompson, an American attorney and activist against obscenity and violence in media and entertainment, would agree with this suggestion. On October 10, Thomas sent A Modest Video Game Proposal, open letter in which he proposes in satirical manner that video games companies stop making ultra-violent video game or be prepare to accept the consequences, to members of the press and to Entertainment Software Association (ESA) president Doug Lowenstein. (Saunderson, 2005) The proposal that video game manufacturers be held accountable is not to suggest that a video make any single person commit a crime; it only acknowledges that fact the video games to contribute to hostile behavior and violent video games promotes real life violence. Danny Ledonne, creator of a Columbine game, does not recognize this fact, despite the fact that Kimveer Gill Columbine-styled shooting in Montreal after Gill played Ledonne's game. Estimating that his game has been downloaded more than 100,000 times, Ledonne says that his game and others are not responsible for the Montreal shootings. "Yeah, there are violent video games, violent movies, but guess what? I can turn on the news and see violence in the real world. Violent video games are a reflection of our societal issues," he said. Ledonne said the intention of the game "was to give some insight into these two young men, to put you into their heads, so to speak, to perhaps understand why they may have done this." (Tsia, 2006)

"Anybody who has enough hate in them to carry out something like this, they would do it with or without the video game." One might expected a statement like that to be made by Ledonne as a way to

deny responsible due the fact that he is a creator of sadistic video game himself. However, for the victims themselves to say such a declaration is little more surprising. Yet the quote was made by Marjorie Lindholm, who as a sophomore at Columbine hid in a room with a lethally wounded teacher. Likewise, Tom Mauser, father of Daniel Mauser, who was murdered at Columbine, said he does not believe video games are the main reason for youth violence. "I think someone has made up their minds. I don't think those games help, and they desensitize," he said. "I think they are aggravating factors. I don't think they take the place of their consciences." (Tsia, 2006)

Another improve that could be made concerning violence is that producers of games should be force to change the content of some ultra-violent video games. This is a reasonable request because there are many studies have proved a connection between video game brutality and the players' aggression in real-life. "Not every single child who plays violent games goes out and shoots people, but personal problems or whatever is going on in their lives may make them vulnerable to this kind of influence," said trauma psychologist Elizabeth Carll, who co-chaired the APA committee behind the resolution. (Tsia, 2006) Producers of these games cannot just simply know that their games cause and promote violence and have that knowledge not affect their production in anyway. That is as irresponsible as an arms dealer selling a convicted murderer a gun.

Along with the help Entertainment Software Rating Board (ESRB), I believe the ESA should enforce my proposals because they are the trade association of the computer and video game industry in the United States. Due to their power and influence, the ESA would be the association that is most probable to turn this paper's suggestions into reality. The advised in this paper system is the best because more parents will stop allowing their little children to play such violent games. Also, companies will stop producing such a great amount of ultra-violent video games when they are held accountable for the after. Combine these benefits and know that in a world where children play less violent video games the world becomes less violent.

Programs and Code

COINS

Marcus Immanuel

Williams

Content

Program #1: Circle which size and color
is dependent on a Jslider

Program #2: Selection Sort

Program #3: Tetrahedron

Program #4: Screen Saver

//s.java

```
import javax.swing.JFrame;  
public class s
```

```
{  
    public static void main(String args[]){  
        SliderFrame sliderFrame = new SliderFrame();  
        sliderFrame.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE);  
        sliderFrame.setSize(280, 340);  
        sliderFrame.setVisible(true);  
    }  
}
```

```
//SliderFrame.java
import java.awt.*;
import javax.swing.*;
import javax.swing.event.*;

import java.awt.geom.*;
import java.util.*;
public class SliderFrame extends JFrame
{
    private JSlider diameterJSlider;
    private OvalPanel myPanel;
    public JTextArea textArea;
    public int dia = 150;

    public SliderFrame()
    {
        super("Slider is rockin' this joint");
        myPanel = new OvalPanel();

        diameterJSlider = new JSlider(SwingConstants.HORIZONTAL, 100, 200,
150 );
        diameterJSlider.setMajorTickSpacing(10);
        diameterJSlider.setBackground(Color.LIGHT_GRAY);
        diameterJSlider.setPaintTicks(true);
        textArea = new JTextArea( "Area:\t"+(Math.PI* ((dia/2)*(dia/2)))+
            ".\nRadius:\t"+dia/2+".\nDiameter:\t"+dia+
            ".\nCircumference:\t"+(2* Math.PI * (dia/2)), 4, 15);
        textArea.setBackground(Color.lightGray);
        diameterJSlider.addChangeListener(
            new ChangeListener()
            {
                public void stateChanged(ChangeEvent e)
                {
                    myPanel.setDiameter( diameterJSlider.getValue());
                    dia = diameterJSlider.getValue();

                    textArea.setText( "Area:\t"+(Math.PI* ((dia/2)*(dia/2)))+
\nRadius:\t"+dia/2+
                        "\nDiameter:\t"+dia+"\nCircumference:\t"+(2* Math.PI *
(dia/2)));
                }
            }
        );

        add(diameterJSlider, BorderLayout.SOUTH);
        add(myPanel, BorderLayout.CENTER);
        add(textArea, BorderLayout.NORTH);
    }
}
```

```
//OvalPanel.java
import java.awt.Graphics;
import java.awt.*;
import javax.swing.*;
import java.awt.geom.*;
import java.util.*;

public class OvalPanel extends JPanel
{
    private int diameter = 150;
    private Color colors[] = { Color.GREEN, Color.CYAN, Color.YELLOW,
        Color.WHITE, Color.RED, Color.ORANGE,
        Color.BLACK, Color.PINK, Color.MAGENTA };
    JPanel textPanel;
    public JTextArea textArea;

    public void paintComponent( Graphics g){

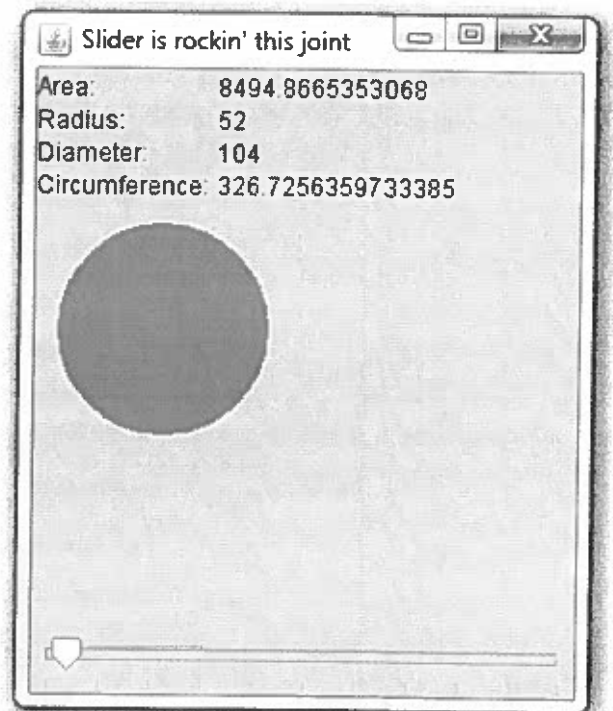
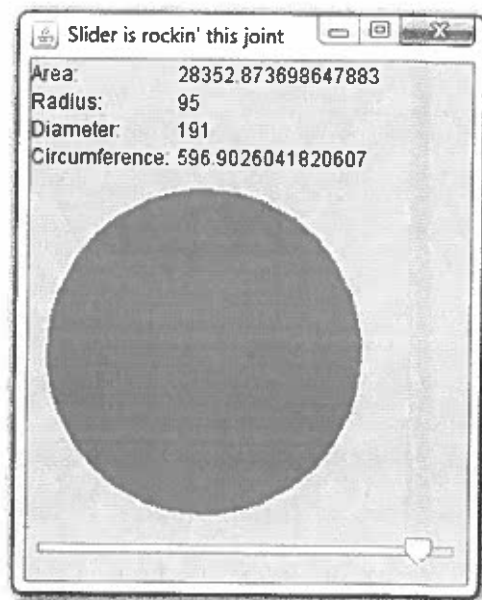
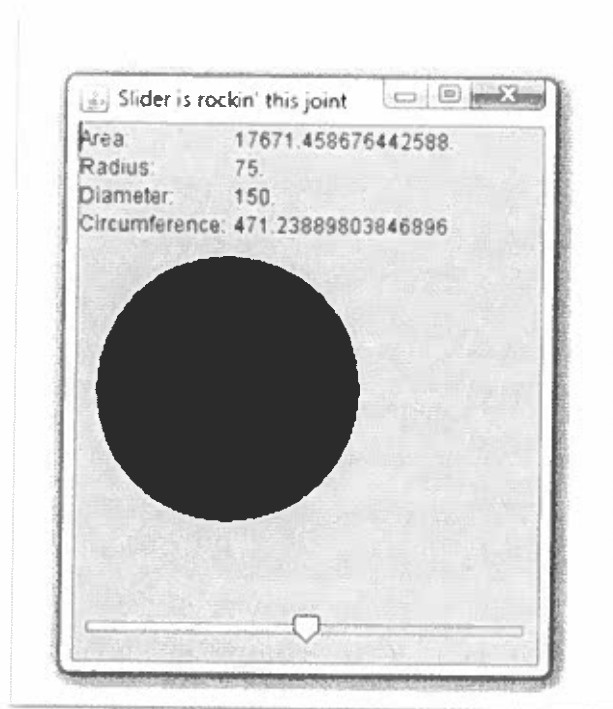
        super .paintComponent(g);
        setBackground( Color.lightGray);
        Random random = new Random();
        Graphics2D g2d = ( Graphics2D ) g;
        int x = random.nextInt( 8 );
        g2d.setColor(colors[x]);
        g2d.fillOval(10,10, diameter, diameter);
        g2d.setStroke(new BasicStroke (x%3));
        x= ((x+1)*3) %8;
        g2d.setColor(colors[x]);
        g2d.draw(new Ellipse2D.Double(10,10, diameter, diameter));
    }

    public void setDiameter(int newDiameter)
    {
        diameter = (newDiameter> 0? newDiameter : 10);
        repaint();
    }

    public Dimension getPreferredSize()
    {
        return new Dimension(200, 200);
    }

    public Dimension getMinimunSize()
    {
        return getPreferredSize();
    }
}
```


Java 2 - Output



```
// Selection.java
import java.util.EmptyStackException;
public class Selection<T>{
    private final int c;
    private int wise;
    private T[] cool;
    public Selection()
    {
        this(3);
    }
    public Selection (int aa)
    {
        c = aa > 0 ? aa : 3;
    }

    public void push( T pushValue )
    {
        if (wise == c -1)
            System.err.print(String.format("WOW", pushValue));
    }
    public T pop()
    {
        if (wise == -1)
            System.err.print("No popping");
        return cool[wise--];
    }
}
```

```
//SelectionSort.java
import java.util.Random;
public class SelectionSort
{
    private int[] data;
    private float[] data2;
    private static Random generator = new Random();

    private Selection<Integer> integerSel;
    private Selection<Float> floatSel;

    public <T extends Comparable<T>> SelectionSort(int size)
    {
        data = new int[size];

        data2 = new float[size];
        for (int i = 0; i < size; i++)
        {
            data[i] = 10 + generator.nextInt(90);
            data2[i] = generator.nextInt(90);
        }

        integerSel = new Selection<Integer>(10);
        floatSel = new Selection<Float>(10);
    }

    public void sort()
    {
        int smallest;
        for( int i = 0; i < data.length - 1; i++)
        {
            smallest = i;
            for (int index = i + 1; index < data.length; index++)
                if (data[index] < data[smallest])
                    smallest = index;
            swap(i, smallest);
            printPass(i+1, smallest);
        }
    }

    public void swap( int first, int second)
    {
        int temporary = data[first];
        data[first] = data[second];
        data[second] = temporary;
    }
}
```

```

    }

    public void printPass( int pass, int index)
    {
        System.out.print(String.format("after pass %2d: ", pass));

        for (int i=0; i < index; i++)
            System.out.print(data[i]+ " ");

        System.out.print( data[index] + "* ");

        for (int i = index + 1; i < data.length; i++ )
            System.out.print(data[i]+" ");

        System.out.print("\n          ");

        for (int j = 0; j < pass; j++)
            System.out.print("-- ");
        System.out.print("\n");

    }

    public String toString()
    {
        StringBuffer temporary = new StringBuffer();

        for (float element : data2)
            temporary.append( element + " ");
        return temporary.toString();
    }

    public static void main ( String[] args)
    {
        SelectionSort sortArray = new SelectionSort(10);
        System.out.println( "Unsorted array");

        System.out.print( sortArray);

        sortArray.sort();

        System.out.println("Sort Array");
        System.out.println(sortArray);

    }
}

```

//SelectionSortTest.java

```
public class SelectionSortTest
{
    public static void main ( String[] args)
    {
        SelectionSort sortArray = new SelectionSort(10);
        System.out.println( "Unsorted array");

        System.out.print( sortArray);

        sortArray.sort();

        System.out.println("Sort Array");
        System.out.println(sortArray);
    }
}
```

Java 1- Output

Unsorted array

75.0 15.0 80.0 39.0 76.0 57.0 23.0 12.0 32.0 42.0

after pass 1: 22 71 35 26 68 68 94* 57 84 25

after pass 2: 22 25 35 26 68 68 94 57 84 71*

after pass 3: 22 25 26 35* 68 68 94 57 84 71

after pass 4: 22 25 26 35* 68 68 94 57 84 71

after pass 5: 22 25 26 35 57 68 94 68* 84 71

after pass 6: 22 25 26 35 57 68* 94 68 84 71

after pass 7: 22 25 26 35 57 68 68 94* 84 71

after pass 8: 22 25 26 35 57 68 68 71 84 94*

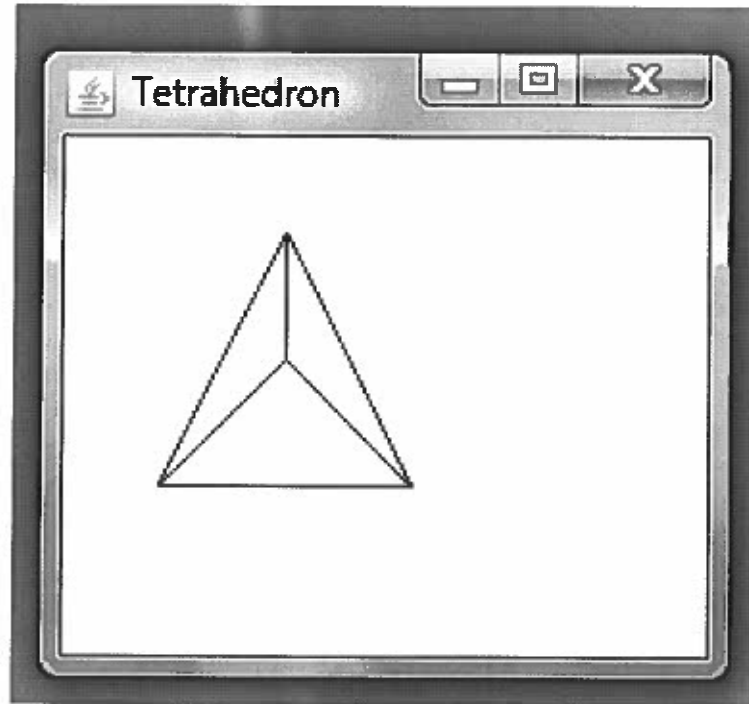
after pass 9: 22 25 26 35 57 68 68 71 84* 94

```
// TetrahedronJPanel.java
// Program draws a tetrahedron.
import java.awt.Color;
import java.awt.geom.GeneralPath;
import java.awt.Graphics;
import java.awt.Graphics2D;
import javax.swing.JPanel;
public class TetrahedronJPanel extends JPanel
{
    // draw tetrahedron
    public void paintComponent( Graphics g )
    {
        super .paintComponent( g );
        /* Define a GeneralPath object that represents a pyramid */
        GeneralPath pyramid = new GeneralPath();
        /* Cast the Graphics reference received by paintComponent to a Graphics2D */
        Graphics2D g2d = ( Graphics2D ) g;
        /* Set the color of the pyramid using Graphics2D method setColor */
        g2d.setColor( new Color(255, 0, 0) );
        /* Use method moveTo to specify the first point in the pyramid */
        int xCord[] = { 70, 30, 70, 70, 110, 30, 110, 70 };
        int yCord[] = { 30, 110, 70, 30, 110, 110, 110, 70 };

        pyramid.moveTo(xCord[0], yCord[0]);

        /* Create a for loop that will access all the points in both arrays
        to create the pyramid in the GeneralPath object */
        for (int count = 1; count < xCord.length; count++)
            pyramid.lineTo( xCord[count], yCord[count]);
        /* Use method closePath to complete the pyramid */
        pyramid.closePath();
        /* Use Graphics2D method draw to display the pyramid */
        g2d.draw(pyramid);
    }
}
```

Java Program #3- Output

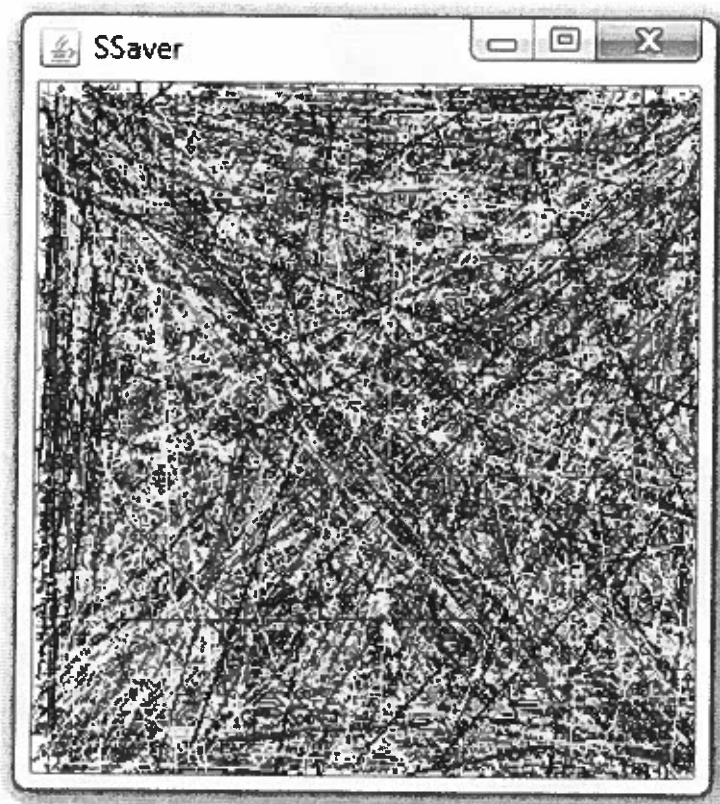
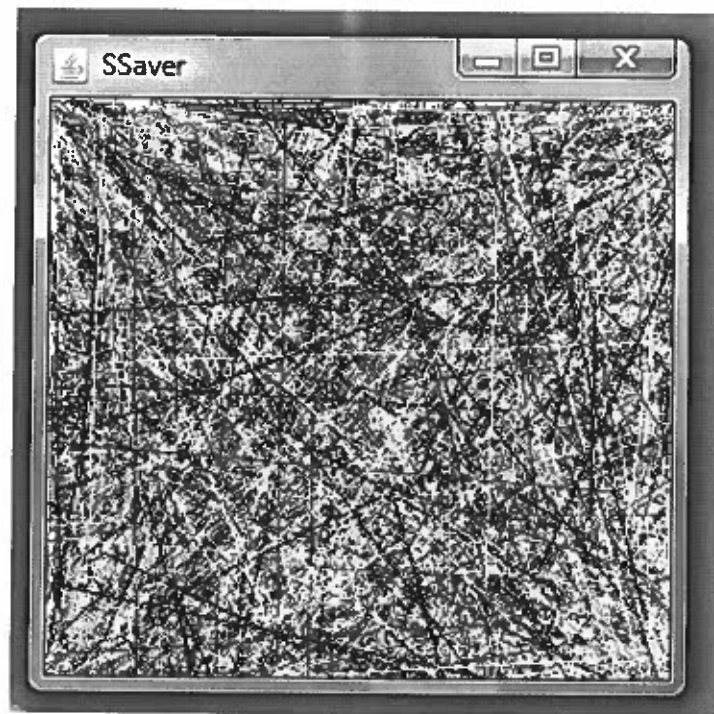



```
// Tetrahedron.java
// Program draws a tetrahedron.
import javax.swing.JFrame;
public class Tetrahedron
{
    public static void main( String args[] )
    {
        // create frame for TetrahedronJPanel
        JFrame frame = new JFrame( "Tetrahedron" );
        frame.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE );
        TetrahedronJPanel tetrahedronJPanel = new TetrahedronJPanel();
        frame.add( tetrahedronJPanel ); // add tetrahedronJPanel to frame
        frame.setSize( 220, 200 ); // set frame size
        frame.setVisible( true ); // display frame
    } // end main
}
```

```
// SSaverPanel.java
// This program draws lines of different colors
import java.awt.Color;
import java.awt.BasicStroke;
import java.awt.geom.Line2D;
import java.awt.Graphics;
import java.awt.Graphics2D;
import java.util.Random;
import javax.swing.JPanel;
import java.awt.color.*;
import java.awt.*;
public class SSaverPanel extends JPanel
{
    private Color colors[] = { Color.GREEN, Color.CYAN, Color.YELLOW, Color.RED,
    Color.ORANGE,
    Color.PINK, Color.MAGENTA, Color.BLUE };
    // create 10 lines
    public void paintComponent( Graphics g )
    {
        super .paintComponent( g );
        setBackground( Color.WHITE ); // set JPanel background
        Random random = new Random(); // get random number generator
        // create 2D by casting g to Graphics 2D
        Graphics2D g2d = ( Graphics2D ) g;
        for (int stupid =1; stupid<100; stupid++)
        {
            for ( int y = 0; y < 100 ; y ++ )
            {
                // choose a random color from array
                /* Randomly choose an element of array colors and pass it to Graphics2D
                method setColor to specify the drawing color */
                int x = random.nextInt( 8);
                g2d.setColor(colors[x]);
                // choose a random thickness from 1-20
                /* Use Graphics2D method setStroke to randomly set the thickness of a line */
                // Choose A random length and draw line
                int t = random.nextInt( 280 );
                int k = random.nextInt( 280 );
                int r = random.nextInt(280 );// this is for length
                int o = random.nextInt( 280 );
                // g2d.setStroke(new BasicStroke (t));//thickness setters
                /* Create a random length line and use Graphics2D method draw
                to display the line */
                g2d.drawLine(t, k, r, o);
            } // end for
        }
        repaint();
    }
    // end method paintComponent
}
```

```
// SSaver2.java
// This program draws SSaver of different colors
import java.awt.Color;
import javax.swing.JFrame;
public class SSaver2
{
    public static void main( String args[] )
    {
        // create frame for SSaverJPanel
        JFrame frame = new JFrame( "SSaver" );
        frame.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE );
        SSaverPanel sSaverPanel = new SSaverPanel();
        frame.add( sSaverPanel ); // add SSaverJPanel to frame
        frame.setSize( 280, 280 ); // set frame size
        frame.setVisible( true ); // display frame
    } // end main
} // end class SSaver2
```

Java Program #4- Output



Presentations

Marcus
Williams

COINS

Kings of Math

**The March 10th 2008 Update of
Marcus Immanuel Williams's Senior Project**

What's is “Kings of Math”?

A computer animation game designed for children in 2nd and 3rd grade. This game is being made using Microsoft C#, XNA and Game Studios.

Kings of Math: What I've Done

Project Implementation: Over the last three weeks, I have refocused the goal of my project because of the questions that the class asked. My games will now be more complexed and be focused at older children. (2nd and 3rd grade)

Kings of Math: What's Next

- Find a way to identify if the selected object was the desire result.
- A playable version of the first game.
- More advanced Controlled Movement and Animation.
- Work with Dr. Lin to design a more complexity layout for my second game.
- Work on collision-prevention

Kings of Math

The Senior Project of
Marcus Immanuel Williams
Spring 2009

Kings Of Math: Introduction

- ◆ Imagine if kids associated math with pleasure instead dull work? What if math was, dare I write it, fun?
- ◆ What if, instead of pencils and papers and erasers, the tools that young students used for solving math problems were virtual balls and flying ships?
- ◆ In short, what if youngster played the mathematics-focused video called 'Kings of Math.'

What is "Kings of Math"?

A computer game with 2D and 3D animation designed for children in 1st - 4th grade. It's goal is to help its player understand mathematics better. This game was using Microsoft C# and XNA's Game Studios.

Kings Of Math: How is it played?

- ◆ Each level of the game will ask the player to solve randomly generated mathematical problems. Depending on the particular level being played, the player will use various inputs devices, range from the mouse to the directional arrow pad.
- ◆ There is no better way to explain it, but to show it to you. So without farther ado, I present to you Kings of Math.

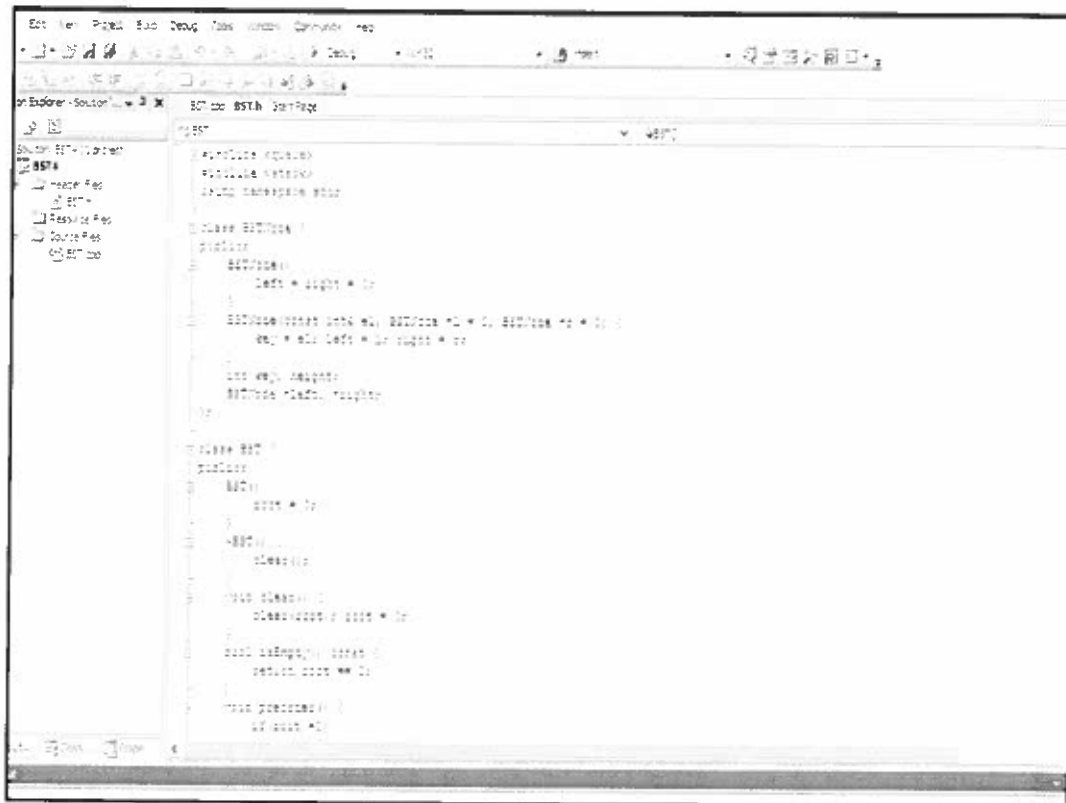
Kings of Math: Conclusion

- ◆ Are there any questions?
- ◆ Are there any comments?
- ◆ Thank for you very much for your time!

COIN 415 Project II: Binary Search Tree

By M. I. Williams

Binary Tree Class



Preorder

```

void BST::preorder(BSTNode *p) {
    if (p != 0) {
        visit(p);
        preorder(p->left);
        preorder(p->right);
    }
}

```

Inorder /Postorder

```
void BST::postorder(BSTNode* p) {
    if (p != 0) {
        postorder(p->left);
        postorder(p->right);
        visit(p);
    }
}
```

```
void BST::inorder(BSTNode *p) {
    if (p != 0) {
        inorder(p->left);
        visit(p);
        inorder(p->right);
    }
}
```

Insertion

```
void BST::insert(const int& el) {
    BSTNode *p = root, *prev = 0;
    while (p != 0) { // Find a place for inserting new Node;
        prev = p;
        if (p->key < el)
            p = p->right;
        else if (p->key > el)
            p = p->left;
        else
            cout<<"That number is already here already!\n\n";
    }
    if (root == 0) // tree is empty;
        root = new BSTNode(el);
    else if (prev->key < el)
        prev->right = new BSTNode(el);
    else prev->left = new BSTNode(el);
}
```

Insertion is pretty much like everyone else, except for it won't enter the same number twice, because I was (and am) unclear about how one would do rotation, AVL with repeated numbers.

Deletion:

search for node

```

void BST::findAndDeleteByCopying(const int& el) {
    BSTNode* p = root, *prev = 0;
    while (p != 0 && p->key != el) {
        prev = p;
        if (p->key < el)
            p = p->right;
        else p = p->left;
    }
    if (p != 0 && p->key == el) {
        if (p == root)
            deleteByCopying(root);
        else if (prev->left == p)
            deleteByCopying(prev->left);
        else deleteByCopying(prev->right);
    } else if (root == 0)
        cout << "The integer '" << el << "' is not in the tree!\n";
    else cout << "Hold up, stat, this tree is like so totally empty!\nInsert a number first!\n\n";
}

```

First it finds what is to be deleted

Deletion:

delete by copying

```
void BST::deleteByCopying(BSTNode* node) {
    BSTNode *previous, *tmp = node;
    cout<<"The integer "<<node->key<<" has been delete...\n\n";
    if (node->right == 0) {
        node = node->left;
    }
    else if (node->left == 0) {
        node = node->right;
    }
    else {
        tmp = node->left;
        previous = node;
        while (tmp->right != 0) {
            previous = tmp;
            tmp = tmp->right;
        }
        node->key = tmp->key;
        if (previous == node)
            previous->left = tmp->left;
        else previous->right = tmp->left;
    }
    delete tmp;
}
```

Now we delete

Rotation(right)

```
if(gr->key<par->key)
    gr->right=child;
else
    gr->left=child;
par->left=child->right;
child->right=par;
```